1) a)

Let $x = \tan u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u \ \mathrm{d}u$

$\therefore \int \frac{x^2}{1+x^2}\mathrm{d}x = \int \frac{\tan^2 u}{1+\tan^2 u} \times \sec^2 u \, \mathrm{d}u$

$= \int \frac{\tan^2 u \sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int \tan^2 u \, \mathrm{d}u$

$= \int (\sec^2 u - 1)\, \mathrm{d}u$

$= \tan u - u + c$

$= x - \arctan x + c$

1) b)

$\frac{x^2}{1+x^2} \equiv \frac{x^2+1-1}{1+x^2} \equiv \frac{x^2+1}{1+x^2} - \frac{1}{1+x^2} \equiv \frac{1+x^2}{1+x^2} - \frac{1}{1+x^2} \equiv 1 - \frac{1}{1+x^2}$

Or using polynomial division:

	1	
x^2	x^2	$0x$
$+0x$	$0x$	-1
$+1$	1	

$\therefore \frac{x^2}{1+x^2} \equiv 1 - \frac{1}{1+x^2}$

Hence:

$\int \frac{x^2}{1+x^2}\mathrm{d}x = \int \left(1 - \frac{1}{1+x^2}\right)\mathrm{d}x = x - \arctan x + c$

2) a)

Let $x = \tan u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u \ \mathrm{d}u$

$\therefore \int \frac{2x^2-1}{1+x^2}\mathrm{d}x = \int \frac{2\tan^2 u-1}{1+\tan^2 u} \times \sec^2 u \, \mathrm{d}u$

$= \int \frac{(2\tan^2 u-1)\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int (2\tan^2 u - 1)\,\mathrm{d}u$

$= \int (2(\sec^2 u - 1) - 1)\,\mathrm{d}u$

$= \int (2\sec^2 u - 3)\,\mathrm{d}u$

$= 2\tan u - 3u + c$

$= 2x - 3\arctan x + c$

Or:

$\frac{2x^2-1}{1+x^2} \equiv \frac{2x^2-1+2-2}{1+x^2} \equiv \frac{2x^2+2}{1+x^2} - \frac{3}{1+x^2} \equiv \frac{2(1+x^2)}{1+x^2} - \frac{3}{1+x^2} \equiv 2 - \frac{3}{1+x^2}$

Or using polynomial division:

	2	
x^2	$2x^2$	$0x$
$+0x$	$0x$	-3
$+1$	2	

$\therefore \frac{2x^2-1}{1+x^2} \equiv 2 - \frac{3}{1+x^2}$

Hence:

$\int \frac{2x^2-1}{1+x^2}\mathrm{d}x = \int \left(2 - \frac{3}{1+x^2}\right)\mathrm{d}x = 2x - 3\arctan x + c$

2) b)

Let $x = \tan u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u \ \mathrm{d}u$

$\therefore \int \frac{5x^2+9}{1+x^2}\mathrm{d}x = \int \frac{5\tan^2 u+9}{1+\tan^2 u} \times \sec^2 u \, \mathrm{d}u$

$= \int \frac{(5\tan^2 u+9)\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int (5\tan^2 u + 9)\,\mathrm{d}u$

$= \int (5(\sec^2 u - 1) + 9)\,\mathrm{d}u$

$= \int (5\sec^2 u + 4)\,\mathrm{d}u$

$= 5\tan u + 4u + c$

$= 5x + 4\arctan x + c$

Or:

$\frac{5x^2+9}{1+x^2} \equiv \frac{5x^2+9+5-5}{1+x^2} \equiv \frac{5x^2+5}{1+x^2} + \frac{4}{1+x^2} \equiv \frac{5(1+x^2)}{1+x^2} + \frac{4}{1+x^2} \equiv 5 + \frac{4}{1+x^2}$

Or using polynomial division:

	5	
x^2	$5x^2$	$0x$
$+0x$	$0x$	$+4$
$+1$	5	

$\therefore \frac{5x^2+9}{1+x^2} \equiv 5 + \frac{4}{1+x^2}$

Hence:

$\int \frac{5x^2+9}{1+x^2}\mathrm{d}x = \int \left(5 + \frac{4}{1+x^2}\right)\mathrm{d}x = 5x + 4\arctan x + c$

3) a)

Use Integration by Substitution:

$\sqrt{x} = \tan u \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(x^{\frac{1}{2}}\right) = \frac{\mathrm{d}}{\mathrm{d}x}(\tan u) \Rightarrow \frac{1}{2}x^{-\frac{1}{2}} = \sec^2 u \frac{\mathrm{d}u}{\mathrm{d}x} \Rightarrow \frac{1}{2\sqrt{x}} = \sec^2 u \frac{\mathrm{d}u}{\mathrm{d}x}$

$\Rightarrow \frac{1}{2\tan u} = \sec^2 u \frac{\mathrm{d}u}{\mathrm{d}x} \Rightarrow \mathrm{d}x = 2\tan u \sec^2 u \, \mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{x}(1+x)}\mathrm{d}x = \int \frac{1}{\tan u(1+\tan^2 u)} \times 2\tan u \sec^2 u \, \mathrm{d}u$

$= \int \frac{2\sec^2 u}{1+\tan^2 u}\mathrm{d}u$

$= \int \frac{2\sec^2 u}{\sec^2 u}\mathrm{d}u$

$= \int 2\,\mathrm{d}u$

$= 2u + c$

$= 2\arctan(\sqrt{x}) + c$

3) b)

Use Integration by Substitution:

$\sqrt{x} = u \Rightarrow \frac{1}{2}x^{-\frac{1}{2}} = \frac{\mathrm{d}u}{\mathrm{d}x} \Rightarrow \frac{1}{2\sqrt{x}} = \frac{\mathrm{d}u}{\mathrm{d}x} \Rightarrow \frac{1}{2u} = \frac{\mathrm{d}u}{\mathrm{d}x} \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

$\therefore \int \frac{1}{\sqrt{x}(1+x)}\mathrm{d}x = \int \frac{1}{u(1+u^2)} \times 2u\,\mathrm{d}u$

$= \int \frac{2}{1+u^2}\mathrm{d}u$

$= 2\arctan u + c$

$= 2\arctan\left(\sqrt{x}\right) + c$

4)

Use Integration by Substitution:

Let $x = \tan u$

$x = \tan u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u\,\mathrm{d}u$

$\therefore \int \frac{1}{(1+x^2)^{\frac{3}{2}}}\mathrm{d}x = \int \frac{1}{(1+\tan^2 u)^{\frac{3}{2}}} \times \sec^2 u\,\mathrm{d}u$

$= \int \frac{\sec^2 u}{(\sec^2 u)^{\frac{3}{2}}}\mathrm{d}u$

$= \int \frac{\sec^2 u}{\sec^3 u}\mathrm{d}u$

$= \int \frac{1}{\sec u}\mathrm{d}u$

$= \int \cos u\,\mathrm{d}u$

$= \sin u + c$

$\tan u = \frac{x}{1} = \frac{opp}{adj}$

$\therefore \int \frac{1}{(1+x^2)^{\frac{3}{2}}}\mathrm{d}x = \frac{x}{\sqrt{1+x^2}} + c$

5)

Use Integration by Substitution:

$u = (x^2 - 1)^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2} \times 2x \times (x^2 - 1)^{-\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{x}{\sqrt{x^2-1}}$

$\Rightarrow \mathrm{d}x = \frac{\sqrt{x^2-1}}{x}\mathrm{d}u \Rightarrow \mathrm{d}x = \frac{u}{x}\mathrm{d}u$

$\therefore \int \frac{1}{x\sqrt{x^2-1}}\mathrm{d}x = \int \frac{1}{xu} \times \frac{u}{x}\mathrm{d}u = \int \frac{1}{x^2}\mathrm{d}u$

$u = (x^2 - 1)^{\frac{1}{2}} \Rightarrow 1 + u^2 = x^2$

$\therefore \int \frac{1}{x^2}\mathrm{d}u = \int \frac{1}{1+u^2}\mathrm{d}u$

$= \arctan u + c$

$= \arctan\left(\sqrt{x^2 - 1}\right) + c$

6) a)

$\cos 2\theta \equiv 1 - 2\sin^2\theta \Rightarrow \sin^2\theta \equiv \frac{1}{2} - \frac{1}{2}\cos 2\theta$

6) b)

Let $x = \sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u \ \mathrm{d}u$

$\therefore \int \frac{x^2}{\sqrt{1-x^2}}\mathrm{d}x = \int \frac{\sin^2 u}{\sqrt{1-\sin^2 u}} \times \cos u \, \mathrm{d}u$

$= \int \frac{\sin^2 u \cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \int \frac{\sin^2 u \cos u}{\cos u}\mathrm{d}u$

$= \int \sin^2 u \, \mathrm{d}u$

$= \int \left(\frac{1}{2} - \frac{1}{2}\cos 2u\right)\mathrm{d}u$

$= \frac{1}{2}u - \frac{1}{4}\sin 2u + c$

$= \frac{1}{2}u - \frac{1}{2}\sin u \cos u + c$

$= \frac{1}{2}\arcsin x - \frac{1}{2}x\sqrt{1 - x^2} + c$

7) a)

Let $2x = \sin u \Rightarrow x = \frac{1}{2}\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\cos u \Rightarrow \mathrm{d}x = \frac{1}{2}\cos u\,\mathrm{d}u$

$\therefore \int \frac{x^2}{\sqrt{1-4x^2}}\mathrm{d}x = \int \frac{x^2}{\sqrt{1-(2x)^2}}\mathrm{d}x = \int \frac{\frac{1}{4}\sin^2 u}{\sqrt{1-\sin^2 u}} \times \frac{1}{2}\cos u\,\mathrm{d}u$

$= \frac{1}{8}\int \frac{\sin^2 u \cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \frac{1}{8}\int \frac{\sin^2 u \cos u}{\cos u}\mathrm{d}u$

$= \frac{1}{8}\int \sin^2 u\,\mathrm{d}u$

$= \frac{1}{8}\int \left(\frac{1}{2} - \frac{1}{2}\cos 2u\right)\mathrm{d}u$

$= \int \left(\frac{1}{16} - \frac{1}{16}\cos 2u\right)\mathrm{d}u$

$= \frac{1}{16}u - \frac{1}{32}\sin 2u + c$

$= \frac{1}{16}u - \frac{1}{16}\sin u \cos u + c$

$= \frac{1}{16}\arcsin 2x - \frac{1}{16} \times 2x \times \sqrt{1-4x^2} + c$

$= \frac{1}{16}\arcsin 2x - \frac{1}{8}x\sqrt{1-4x^2} + c$

7) b)

Let $\frac{1}{2}x = \sin u \Rightarrow x = 2\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\cos u \Rightarrow \mathrm{d}x = 2\cos u\,\mathrm{d}u$

$\therefore \int \frac{x^2}{\sqrt{4-x^2}}\mathrm{d}x = \int \frac{x^2}{\sqrt{4\left(1-\frac{1}{4}x^2\right)}}\mathrm{d}x = \frac{1}{2}\int \frac{x^2}{\sqrt{1-\left(\frac{1}{2}x\right)^2}}\mathrm{d}x = \frac{1}{2}\int \frac{4\sin^2 u}{\sqrt{1-\sin^2 u}} \times 2\cos u\,\mathrm{d}u$

$= 4\int \frac{\sin^2 u \cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= 4\int \frac{\sin^2 u \cos u}{\cos u}\mathrm{d}u$

$= 4\int \sin^2 u\,\mathrm{d}u$

$= 4\int \left(\frac{1}{2} - \frac{1}{2}\cos 2u\right)\mathrm{d}u$

$= \int (2 - 2\cos 2u)\,\mathrm{d}u$

$= 2u - \sin 2u + c$

$= 2u - 2\sin u \cos u + c$

$= 2\arcsin\left(\frac{1}{2}x\right) - 2 \times \frac{1}{2}x \times \sqrt{1-\frac{1}{4}x^2} + c$

$= 2\arcsin\left(\frac{1}{2}x\right) - x\sqrt{1-\frac{1}{4}x^2} + c$

7) c)

Let $\frac{1}{\sqrt{2}}x = \sin u \Rightarrow x = \sqrt{2}\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sqrt{2}\cos u \Rightarrow \mathrm{d}x = \sqrt{2}\cos u\,\mathrm{d}u$

$\therefore \int \frac{x^2}{\sqrt{4-2x^2}}\mathrm{d}x = \int \frac{x^2}{\sqrt{4\left(1-\frac{1}{2}x^2\right)}}\mathrm{d}x = \frac{1}{2}\int \frac{x^2}{\sqrt{1-\left(\frac{1}{\sqrt{2}}x\right)^2}}\mathrm{d}x = \frac{1}{2}\int \frac{2\sin^2 u}{\sqrt{1-\sin^2 u}} \times \sqrt{2}\cos u\,\mathrm{d}u$

$= \sqrt{2}\int \frac{\sin^2 u \cos u}{\sqrt{\cos^2 u}}\mathrm{d}u$

$= \sqrt{2}\int \frac{\sin^2 u \cos u}{\cos u}\mathrm{d}u$

$= \sqrt{2}\int \sin^2 u\,\mathrm{d}u$

$= \sqrt{2}\int \left(\frac{1}{2} - \frac{1}{2}\cos 2u\right)\mathrm{d}u$

$= \int \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\cos 2u\right)\mathrm{d}u$

$= \frac{\sqrt{2}}{2}u - \frac{\sqrt{2}}{4}\sin 2u + c$

$= \frac{\sqrt{2}}{2}u - \frac{\sqrt{2}}{2}\sin u \cos u + c$

$= \frac{\sqrt{2}}{2}\arcsin\left(\frac{1}{\sqrt{2}}x\right) - \frac{\sqrt{2}}{2} \times \frac{1}{\sqrt{2}}x \times \sqrt{1-\frac{1}{2}x^2} + c$

$= \frac{\sqrt{2}}{2}\arcsin\left(\frac{1}{\sqrt{2}}x\right) - \frac{1}{2}x\sqrt{1-\frac{1}{2}x^2} + c$

8) a)

Let $x = \sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u\ \mathrm{d}u$

$\therefore \int \frac{2x^2+3}{\sqrt{1-x^2}}\mathrm{d}x = \int \left(\frac{2x^2}{\sqrt{1-x^2}} + \frac{3}{\sqrt{1-x^2}}\right)\mathrm{d}x = \int \left(\frac{2\sin^2 u}{\sqrt{1-\sin^2 u}} + \frac{3}{\sqrt{1-\sin^2 u}}\right) \times \cos u\,\mathrm{d}u$

$= \int \left(\frac{2\sin^2 u \cos u}{\sqrt{\cos^2 u}} + \frac{3\cos u}{\sqrt{\cos^2 u}}\right)\mathrm{d}u$

$= \int \left(\frac{2\sin^2 u \cos u}{\cos u} + \frac{3\cos u}{\cos u}\right)\mathrm{d}u$

$= \int (2\sin^2 u + 3)\,\mathrm{d}u$

$= \int \left(2\left(\frac{1}{2} - \frac{1}{2}\cos 2u\right) + 3\right)\mathrm{d}u$

$= \int (4 - \cos 2u)\,\mathrm{d}u$

$= 4u - \frac{1}{2}\sin 2u + c$

$= 4u - \sin u \cos u + c$

$= 4\arcsin x - x\sqrt{1-x^2} + c$

8) b)

Let $3x = \sin u \Rightarrow x = \frac{1}{3}\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{3}\cos u \Rightarrow \mathrm{d}x = \frac{1}{3}\cos u \ \mathrm{d}u$

$\therefore \int \frac{3-x^2}{\sqrt{1-9x^2}}\mathrm{d}x = \int \frac{3-x^2}{\sqrt{1-(3x)^2}}\mathrm{d}x = \int \left(\frac{3}{\sqrt{1-(3x)^2}} - \frac{x^2}{\sqrt{1-(3x)^2}}\right)\mathrm{d}x$

$= \int \left(\frac{3}{\sqrt{1-\sin^2 u}} - \frac{\frac{1}{9}\sin^2 u}{\sqrt{1-\sin^2 u}}\right) \times \frac{1}{3}\cos u \, \mathrm{d}u$

$= \int \left(\frac{\cos u}{\sqrt{\cos^2 u}} - \frac{\frac{1}{27}\sin^2 u \cos u}{\sqrt{\cos^2 u}}\right)\mathrm{d}u$

$= \int \left(\frac{\cos u}{\cos u} - \frac{\frac{1}{27}\sin^2 u \cos u}{\cos u}\right)\mathrm{d}u$

$= \int \left(1 - \frac{1}{27}\sin^2 u\right)\mathrm{d}u$

$= \int \left(1 - \frac{1}{27}\left(\frac{1}{2} - \frac{1}{2}\cos 2u\right)\right)\mathrm{d}u$

$= \int \left(\frac{53}{54} + \frac{1}{54}\cos 2u\right)\mathrm{d}u$

$= \frac{53}{54}u + \frac{1}{108}\sin 2u + c$

$= \frac{53}{54}u + \frac{1}{54}\sin u \cos u + c$

$= \frac{53}{54}\arcsin 3x + \frac{1}{54} \times 3x \times \sqrt{1-9x^2} + c$

$= \frac{53}{54}\arcsin 3x + \frac{1}{18}x\sqrt{1-9x^2} + c$

8) c)

Let $\sqrt{3}x = \sin u \Rightarrow x = \frac{1}{\sqrt{3}}\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{\sqrt{3}}\cos u \Rightarrow \mathrm{d}x = \frac{1}{\sqrt{3}}\cos u \ \mathrm{d}u$

$\therefore \int \frac{3x^2+1}{\sqrt{1-3x^2}}\mathrm{d}x = \int \frac{3x^2+1}{\sqrt{1-(\sqrt{3}x)^2}}\mathrm{d}x = \int \left(\frac{3x^2}{\sqrt{1-(\sqrt{3}x)^2}} + \frac{1}{\sqrt{1-(\sqrt{3}x)^2}}\right)\mathrm{d}x$

$= \int \left(\frac{\sin^2 u}{\sqrt{1-\sin^2 u}} + \frac{1}{\sqrt{1-\sin^2 u}}\right) \times \frac{1}{\sqrt{3}}\cos u \, \mathrm{d}u$

$= \int \left(\frac{\sin^2 u}{\sqrt{\cos^2 u}} + \frac{1}{\sqrt{\cos^2 u}}\right) \times \frac{1}{\sqrt{3}}\cos u \, \mathrm{d}u$

$= \int \left(\frac{\sin^2 u}{\cos u} + \frac{1}{\cos u}\right) \times \frac{1}{\sqrt{3}}\cos u \, \mathrm{d}u$

$= \int \left(\frac{1}{\sqrt{3}}\sin^2 u + \frac{1}{\sqrt{3}}\right)\mathrm{d}u$

$= \int \left(\frac{1}{\sqrt{3}}\left(\frac{1}{2} - \frac{1}{2}\cos 2u\right) + \frac{1}{\sqrt{3}}\right)\mathrm{d}u$

$= \int \left(\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{6}\cos 2u\right)\mathrm{d}u$

$= \frac{\sqrt{3}}{2}u - \frac{\sqrt{3}}{12}\sin 2u + c$

$= \frac{\sqrt{3}}{2}u - \frac{\sqrt{3}}{6}\sin u \cos u + c$

$= \frac{\sqrt{3}}{2}\arcsin\sqrt{3}x - \frac{\sqrt{3}}{6} \times \sqrt{3}x \times \sqrt{1-3x^2} + c$

$= \frac{\sqrt{3}}{2}\arcsin\sqrt{3}x - \frac{1}{2}x\sqrt{1-3x^2} + c$

9)

$$\int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{8+8x^2}{\sqrt{1-4x^2}}\,\mathrm{d}x = \int_{-\frac{1}{4}}^{\frac{1}{4}} \left(\frac{8}{\sqrt{1-4x^2}} + \frac{8x^2}{\sqrt{1-4x^2}}\right)\mathrm{d}x$$

$$= \int_{-\frac{1}{4}}^{\frac{1}{4}} \left(\frac{8}{\sqrt{1-(2x)^2}} + \frac{8x^2}{\sqrt{1-4x^2}}\right)\mathrm{d}x$$

Consider $\int \frac{8x^2}{\sqrt{1-4x^2}}\,\mathrm{d}x$

Use Integration by Substitution

Let $2x = \sin u \Rightarrow x = \frac{1}{2}\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\cos u \Rightarrow \mathrm{d}x = \frac{1}{2}\cos u\,\mathrm{d}u$

$$\int \frac{8x^2}{\sqrt{1-4x^2}}\,\mathrm{d}x = \int \frac{2\sin^2 u}{\sqrt{1-\sin^2 u}} \times \frac{1}{2}\cos u\,\mathrm{d}u$$

$$= \int \frac{\sin^2 u \cos u}{\sqrt{\cos^2 u}}\,\mathrm{d}u$$

$$= \int \frac{\sin^2 u \cos u}{\cos u}\,\mathrm{d}u$$

$$= \int \sin^2 u\,\mathrm{d}u$$

$$\cos 2x = 1 - 2\sin^2 x \Rightarrow \sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$$

$$\therefore \int \sin^2 u\,\mathrm{d}u = \int \left(\frac{1}{2} - \frac{1}{2}\cos 2u\right)\mathrm{d}u$$

$$= \frac{1}{2}u - \frac{1}{4}\sin 2u + c$$

$$= \frac{1}{2}u - \frac{1}{2}\sin u \cos u + c$$

$$= \frac{1}{2}u - \frac{1}{2}\sin u \sqrt{1-\sin^2 u} + c$$

$$= \frac{1}{2}\arcsin 2x - x\sqrt{1-4x^2} + c$$

So:

$$\int_{-\frac{1}{4}}^{\frac{1}{4}} \left(\frac{8}{\sqrt{1-(2x)^2}} + \frac{8x^2}{\sqrt{1-4x^2}}\right)\mathrm{d}x$$

$$= \left[8 \times \frac{1}{2} \times \arcsin 2x + \frac{1}{2}\arcsin 2x - x\sqrt{1-4x^2}\right]_{-\frac{1}{4}}^{\frac{1}{4}}$$

$$= \left[\frac{9}{2}\arcsin 2x - x\sqrt{1-4x^2}\right]_{-\frac{1}{4}}^{\frac{1}{4}}$$

$$= \left(\frac{9}{2}\arcsin\frac{1}{2} - \frac{1}{4} \times \frac{\sqrt{3}}{2}\right) - \left(\frac{9}{2}\arcsin\left(-\frac{1}{2}\right) - \left(-\frac{1}{4}\right) \times \frac{\sqrt{3}}{2}\right)$$

$$= \left(\frac{9}{2} \times \frac{\pi}{6} - \frac{\sqrt{3}}{8}\right) - \left(\frac{9}{2} \times \left(-\frac{\pi}{6}\right) + \frac{\sqrt{3}}{8}\right)$$

$$= \left(\frac{3\pi}{4} - \frac{\sqrt{3}}{8}\right) - \left(-\frac{3\pi}{4} + \frac{\sqrt{3}}{8}\right)$$

$$= \frac{3\pi}{2} - \frac{\sqrt{3}}{4}$$

10) a)

$\frac{\mathrm{d}}{\mathrm{d}x}\left(3x\sqrt{1-9x^2}-\arccos 3x\right)$

$=3x\times\frac{1}{2}(-18x)(1-9x^2)^{-\frac{1}{2}}+3(1-9x^2)^{\frac{1}{2}}+\frac{3}{\sqrt{1-(3x)^2}}$

$=-27x^2(1-9x^2)^{-\frac{1}{2}}+3(1-9x^2)^{\frac{1}{2}}+\frac{3}{\sqrt{1-9x^2}}$

$=-\frac{27x^2}{\sqrt{1-9x^2}}+3\sqrt{1-9x^2}+\frac{3}{\sqrt{1-9x^2}}$

$=3\sqrt{1-9x^2}+\frac{3-27x^2}{\sqrt{1-9x^2}}$

$=3\sqrt{1-9x^2}+\frac{3(1-9x^2)}{\sqrt{1-9x^2}}$

$=3\sqrt{1-9x^2}+3\sqrt{1-9x^2}$

$=6\sqrt{1-9x^2}$

So $k=6$

10) b)

$\int_{\frac{1}{6}}^{\frac{\sqrt{3}}{6}}\sqrt{1-9x^2}\,\mathrm{d}x=\frac{1}{6}\left[3x\sqrt{1-9x^2}-\arccos 3x\right]_{\frac{1}{6}}^{\frac{\sqrt{3}}{6}}$

$=\frac{1}{6}\left(\left(3\times\frac{\sqrt{3}}{6}\times\sqrt{1-9\left(\frac{\sqrt{3}}{6}\right)^2}-\arccos\left(3\times\frac{\sqrt{3}}{6}\right)\right)-\left(3\times\frac{1}{6}\times\sqrt{1-9\left(\frac{1}{6}\right)^2}-\right.\right.$

$\left.\left.\arccos\left(3\times\frac{1}{6}\right)\right)\right)$

$=\frac{1}{6}\left(\left(3\times\frac{\sqrt{3}}{6}\times\frac{1}{2}-\arccos\left(\frac{\sqrt{3}}{2}\right)\right)-\left(3\times\frac{1}{6}\times\frac{\sqrt{3}}{2}-\arccos\left(\frac{1}{2}\right)\right)\right)$

$=\frac{1}{6}\left(\left(\frac{\sqrt{3}}{4}-\frac{\pi}{6}\right)-\left(\frac{\sqrt{3}}{4}-\frac{\pi}{3}\right)\right)$

$=\frac{1}{6}\left(-\frac{\pi}{6}+\frac{\pi}{3}\right)$

$=\frac{\pi}{36}$

11) a)

$\frac{\mathrm{d}}{\mathrm{d}x}\left(2(x-5)\sqrt{11+10x-x^2}+72\arcsin\left(\frac{x-5}{6}\right)\right)$

$=2(x-5)\times\frac{1}{2}(10-2x)(11+10x-x^2)^{-\frac{1}{2}}+2(11+10x-x^2)^{\frac{1}{2}}+72\times\frac{1}{6}\times\frac{1}{\sqrt{1-\left(\frac{x-5}{6}\right)^2}}$

$=-2(x-5)^2(11+10x-x^2)^{-\frac{1}{2}}+2(11+10x-x^2)^{\frac{1}{2}}+\frac{12}{\sqrt{1-\frac{x^2-10x+25}{36}}}$

$=-\frac{2(x-5)^2}{\sqrt{11+10x-x^2}}+2(11+10x-x^2)^{\frac{1}{2}}+\frac{12}{\sqrt{\frac{1}{36}(36-(x^2-10x+25))}}$

$=-\frac{2(x-5)^2}{\sqrt{11+10x-x^2}}+2(11+10x-x^2)^{\frac{1}{2}}+\frac{72}{\sqrt{11+10x-x^2}}$

$=2(11+10x-x^2)^{\frac{1}{2}}+\frac{72-2(x-5)^2}{\sqrt{11+10x-x^2}}$

$=2(11+10x-x^2)^{\frac{1}{2}}+\frac{72-2(x^2-10x+25)}{\sqrt{11+10x-x^2}}$

$=2(11+10x-x^2)^{\frac{1}{2}}+\frac{2(11+10x-x^2)}{\sqrt{11+10x-x^2}}$

$=2(11+10x-x^2)^{\frac{1}{2}}+2(11+10x-x^2)^{\frac{1}{2}}$

$=4\sqrt{11+10x-x^2}$

$\therefore k=4$

11) b)

$\int_5^8\sqrt{11+10x-x^2}\,\mathrm{d}x=\left[\frac{1}{2}(x-5)\sqrt{11+10x-x^2}+18\arcsin\left(\frac{x-5}{6}\right)\right]_5^8$

$=\left(\frac{1}{2}\times3\times\sqrt{27}+18\arcsin\frac{1}{2}\right)-\left(\frac{1}{2}\times0\times\sqrt{36}+18\arcsin0\right)$

$=\left(\frac{1}{2}\times3\times3\sqrt{3}+3\pi\right)-(0+0)$

$=\frac{9\sqrt{3}}{2}+3\pi$

12) a)

$19+18x-x^2$

$\equiv19-(x^2-18x)$

$\equiv19-((x-9)^2-81)$

$\equiv19-(x-9)^2+81$

$\equiv100-(x-9)^2$

So $p=100$ and $q=-9$

12) b)

Use Integration by Substitution:

Let $\frac{x-9}{10} = \sin u \Rightarrow x = 10\sin u + 9 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 10\cos u \Rightarrow \mathrm{d}x = 10\cos u\,\mathrm{d}u$

$\int \sqrt{19 + 18x - x^2}\,\mathrm{d}x$

$= \int \sqrt{100 - (x-9)^2}\,\mathrm{d}x$

$= \int \sqrt{100\left(1 - \frac{1}{100}(x-9)^2\right)}\,\mathrm{d}x$

$= 10\int \sqrt{1 - \left(\frac{x-9}{10}\right)^2}\,\mathrm{d}x$

$= 10\int \sqrt{1 - \sin^2 u} \times 10\cos u\,\mathrm{d}u$

$= 100\int \sqrt{\cos^2 u} \times \cos u\,\mathrm{d}u$

$= 100\int \cos u \times \cos u\,\mathrm{d}u$

$= 100\int \cos^2 u\,\mathrm{d}u$

$= 100\int \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)\mathrm{d}u$

$= 100\left(\frac{1}{4}\sin 2u + \frac{1}{2}u\right) + c$

$= 25\sin 2u + 50u + c$

$= 50\sin u\cos u + 50u + c$

$= 50\sin u\sqrt{1 - \sin^2 u} + 50u + c$

$= 50\left(\frac{x-9}{10}\right)\sqrt{1 - \left(\frac{x-9}{10}\right)^2} + 50\arcsin\left(\frac{x-9}{10}\right) + c$

$= 5(x-9)\sqrt{\frac{1}{100}(100 - (x-9)^2)} + 50\arcsin\left(\frac{x-9}{10}\right) + c$

$= \frac{1}{2}(x-9)\sqrt{100 - (x-9)^2} + 50\arcsin\left(\frac{x-9}{10}\right) + c$

$= \frac{1}{2}(x-9)\sqrt{19 + 19x - x^2} + 50\arcsin\left(\frac{x-9}{10}\right) + c$

13)

$3 + 2x - x^2$

$\equiv 3 - (x^2 - 2x)$

$\equiv 3 - ((x-1)^2 - 1)$

$\equiv 3 - (x-1)^2 + 1$

$\equiv 4 - (x-1)^2$

Use Integration by Substitution:

Let $\frac{x-1}{2} = \sin u \Rightarrow x = 2\sin u + 1 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\cos u \Rightarrow \mathrm{d}x = 2\cos u\,\mathrm{d}u$

$\int \sqrt{3 + 2x - x^2}\,\mathrm{d}x$

$= \int \sqrt{4 - (x-1)^2}\,\mathrm{d}x$

$= \int \sqrt{4\left(1 - \frac{1}{4}(x-1)^2\right)}\,\mathrm{d}x$

$= 2\int \sqrt{1 - \left(\frac{x-1}{2}\right)^2}\,\mathrm{d}x$

$= 2\int \sqrt{1 - \sin^2 u} \times 2\cos u\,\mathrm{d}u$

$= 4\int \sqrt{\cos^2 u} \times \cos u\,\mathrm{d}u$

$= 4\int \cos u \times \cos u\,\mathrm{d}u$

$= 4\int \cos^2 u\,\mathrm{d}u$

$= 4\int \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)\mathrm{d}u$

$= 4\left(\frac{1}{4}\sin 2u + \frac{1}{2}u\right) + c$

$= \sin 2u + 2u + c$

$= 2\sin u \cos u + 2u + c$

$= 2\sin u\sqrt{1 - \sin^2 u} + 2u + c$

$= 2\left(\frac{x-1}{2}\right)\sqrt{1 - \left(\frac{x-1}{2}\right)^2} + 2\arcsin\left(\frac{x-1}{2}\right) + c$

$= (x-1)\sqrt{\frac{1}{4}(4 - (x-1)^2)} + 2\arcsin\left(\frac{x-1}{2}\right) + c$

$= \frac{1}{2}(x-1)\sqrt{4 - (x-1)^2} + 2\arcsin\left(\frac{x-1}{2}\right) + c$

$= \frac{1}{2}(x-1)\sqrt{3 + 2x - x^2} + 2\arcsin\left(\frac{x-1}{2}\right) + c$

14) a)

$\sin 3A \equiv \sin(2A + A) \equiv \sin 2A \cos A + \cos 2A \sin A$

$\equiv 2 \sin A \cos A \times \cos A + (1 - 2 \sin^2 A) \sin A$

$\equiv 2 \sin A \cos^2 A + \sin A - 2 \sin^3 A$

$\equiv 2 \sin A (1 - \sin^2 A) + \sin A - 2 \sin^3 A$

$\equiv 2 \sin A - 2 \sin^3 A + \sin A - 2 \sin^3 A$

$\equiv 3 \sin A - 4 \sin^3 A$

Alternatively:

$(\cos A + \mathrm{i} \sin A)^3 \equiv \cos 3A + \mathrm{i} \sin 3A$

$\Rightarrow \cos^3 A + 3\mathrm{i} \cos^2 A \sin A - 3 \cos A \sin^2 A - \mathrm{i} \sin^3 A \equiv \cos 3A + \mathrm{i} \sin 3A$

Im: $\sin 3A \equiv 3 \cos^2 A \sin A - \sin^3 A$

$\equiv 3(1 - \sin^2 A) \sin A - \sin^3 A$

$\equiv 3 \sin A - 3 \sin^3 A - \sin^3 A$

$\equiv 3 \sin A - 4 \sin^3 A$

14) b)

$\cos 3A \equiv \cos(2A + A) \equiv \cos 2A \cos A - \sin 2A \sin A$

$\equiv (2 \cos^2 A - 1) \cos A - (2 \sin A \cos A) \sin A$

$\equiv 2 \cos^3 A - \cos A - 2 \sin^2 A \cos A$

$\equiv 2 \cos^3 A - \cos A - 2(1 - \cos^2 A) \cos A$

$\equiv 2 \cos^3 A - \cos A - 2 \cos A + 2 \cos^3 A$

$\equiv 4 \cos^3 A - 3 \cos A$

Alternatively:

$(\cos A + \mathrm{i} \sin A)^3 \equiv \cos 3A + \mathrm{i} \sin 3A$

$\Rightarrow \cos^3 A + 3\mathrm{i} \cos^2 A \sin A - 3 \cos A \sin^2 A - \mathrm{i} \sin^3 A \equiv \cos 3A + \mathrm{i} \sin 3A$

Re: $\cos 3A \equiv \cos^3 A - 3 \cos A \sin^2 A$

$\equiv \cos^3 A - 3 \cos A (1 - \cos^2 A)$

$\equiv \cos^3 A - 3 \cos A + 3 \cos^3 A$

$\equiv 4 \cos^3 A - 3 \cos A$

14) c)

Use Integration by Substitution:

Let $x = \sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u \, \mathrm{d}u$

$I = \int \frac{6x^3 - 4x + 1}{\sqrt{1 - x^2}} \mathrm{d}x = \int \frac{6 \sin^3 u - 4 \sin u + 1}{\sqrt{1 - \sin^2 u}} \times \cos u \, \mathrm{d}u$

$= \int \frac{6\sin^3 u - 4\sin u + 1}{\sqrt{\cos^2 u}} \times \cos u \, \mathrm{d}u$

$= \int \frac{6\sin^3 u - 4\sin u + 1}{\cos u} \times \cos u \, \mathrm{d}u$

$= \int (6\sin^3 u - 4\sin u + 1) \, \mathrm{d}u$

$\sin 3u \equiv 3\sin u - 4\sin^3 u \Rightarrow \sin^3 u \equiv \frac{3}{4}\sin u - \frac{1}{4}\sin 3u$

$I = \int \left(6\left(\frac{3}{4}\sin u - \frac{1}{4}\sin 3u\right) - 4\sin u + 1\right) \mathrm{d}u$

$= \int \left(\frac{9}{2}\sin u - \frac{3}{2}\sin 3u - 4\sin u + 1\right) \mathrm{d}u$

$= \int \left(\frac{1}{2}\sin u - \frac{3}{2}\sin 3u + 1\right) \mathrm{d}u$

$= -\frac{1}{2}\cos u + \frac{1}{2}\cos 3u + u + c$

$\cos 3u \equiv 4\cos^3 u - 3\cos u$

$I = -\frac{1}{2}\cos u + \frac{1}{2}(4\cos^3 u - 3\cos u) + u + c$

$= -\frac{1}{2}\cos u + 2\cos^3 u - \frac{3}{2}\cos u + u + c$

$= -2\cos u + 2\cos^3 u + u + c$

$= -2\sqrt{1 - \sin^2 u} + 2(1 - \sin^2 u)^{\frac{3}{2}} + u + c$

$= -2\sqrt{1 - x^2} + 2(1 - x^2)^{\frac{3}{2}} + \arcsin x + c$

Alternatively:

Use Integration by Substitution:

Let $x = \sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u \, \mathrm{d}u$

$I = \int \frac{6x^3 - 4x + 1}{\sqrt{1 - x^2}} \mathrm{d}x = \int \frac{6\sin^3 u - 4\sin u + 1}{\sqrt{1 - \sin^2 u}} \times \cos u \, \mathrm{d}u$

$= \int \frac{6\sin^3 u - 4\sin u + 1}{\sqrt{\cos^2 u}} \times \cos u \, \mathrm{d}u$

$= \int \frac{6\sin^3 u - 4\sin u + 1}{\cos u} \times \cos u \, \mathrm{d}u$

$= \int (6\sin^3 u - 4\sin u + 1) \, \mathrm{d}u$

$= \int (6\sin u \times \sin^2 u - 4\sin u + 1) \, \mathrm{d}u$

$= \int (6\sin u \, (1 - \cos^2 u) - 4\sin u + 1) \, \mathrm{d}u$

$= \int (6\sin u - 6\sin u \cos^2 u - 4\sin u + 1) \, \mathrm{d}u$

$= \int (2\sin u - 6\sin u \cos^2 u + 1) \, \mathrm{d}u$

$= -2\cos u + 2\cos^3 u + u + c$

$= -2\sqrt{1 - \sin^2 u} + 2(1 - \sin^2 u)^{\frac{3}{2}} + u + c$

$= -2\sqrt{1 - x^2} + 2(1 - x^2)^{\frac{3}{2}} + \arcsin x + c$

15) a)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\arctan\left(\sqrt{\frac{x}{3}}\right)\right)=\frac{\mathrm{d}}{\mathrm{d}x}\left(\arctan\left(\frac{x^{\frac{1}{2}}}{\sqrt{3}}\right)\right)$$

$$=\frac{1}{\sqrt{3}}\times\frac{1}{2}x^{-\frac{1}{2}}\times\frac{1}{1+\left(\frac{x^{\frac{1}{2}}}{\sqrt{3}}\right)^2}$$

$$=\frac{1}{2\sqrt{3}x^{\frac{1}{2}}\left(1+\frac{x}{3}\right)}$$

$$=\frac{3}{2\sqrt{3}x^{\frac{1}{2}}(3+x)}$$

$$=\frac{\sqrt{3}}{2x^{\frac{1}{2}}(3+x)}$$

15) b)

$$\frac{\sqrt{3}}{2x^{\frac{1}{2}}(3+x)}=\frac{\sqrt{3}x^{\frac{1}{2}}}{2x(3+x)}$$

$$=\frac{\sqrt{3}}{2}\times\frac{x^{\frac{1}{2}}}{x(3+x)}$$

$$=\frac{\sqrt{3}}{2}\times\frac{\sqrt{x}}{3x+x^2}$$

$$\therefore\int_1^3\frac{\sqrt{x}}{3x+x^2}\,\mathrm{d}x=\frac{1}{\frac{\sqrt{3}}{2}}\int_1^3\frac{\sqrt{3}}{2x^{\frac{1}{2}}(3+x)}\,\mathrm{d}x$$

$$=\frac{2\sqrt{3}}{3}\int_1^3\frac{\sqrt{3}}{2x^{\frac{1}{2}}(3+x)}\,\mathrm{d}x$$

$$=\frac{2\sqrt{3}}{3}\left[\arctan\left(\sqrt{\frac{x}{3}}\right)\right]_1^3$$

$$=\frac{2\sqrt{3}}{3}\left(\arctan\left(\sqrt{\frac{3}{3}}\right)-\arctan\left(\sqrt{\frac{1}{3}}\right)\right)$$

$$=\frac{2\sqrt{3}}{3}\left(\arctan 1-\arctan\frac{1}{\sqrt{3}}\right)$$

$$=\frac{2\sqrt{3}}{3}\left(\frac{\pi}{4}-\frac{\pi}{6}\right)$$

$$=\frac{2\sqrt{3}}{3}\times\frac{\pi}{12}$$

$$=\frac{\pi\sqrt{3}}{18}$$

$$=\frac{3\pi}{18\sqrt{3}}$$

$$=\frac{\pi}{6\sqrt{3}}$$

16)

Use Integration by Substitution:

$u^2 = x - 3 \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(u^2) = \frac{\mathrm{d}}{\mathrm{d}x}(x-3) \Rightarrow 2u\frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

$x + 6 = u^2 + 9$

$\int \frac{\sqrt{x-3}}{x+6}\mathrm{d}x = \int \frac{\sqrt{u^2}}{u^2+9} \times 2u\,\mathrm{d}u = \int \frac{2u^2}{9+u^2}\mathrm{d}u$

$\frac{2u^2}{9+u^2} \equiv \frac{2u^2+18-18}{9+u^2} \equiv \frac{2u^2+18}{9+u^2} - \frac{18}{9+u^2} \equiv \frac{2(9+u^2)}{9+u^2} - \frac{18}{9+u^2} \equiv 2 - \frac{18}{9+u^2}$

Or using polynomial division:

	2	
u^2	$2u^2$	$0u$
$+0u$	$0u$	-18
$+9$	18	

$\therefore \frac{2u^2}{9+u^2} \equiv 2 - \frac{18}{9+u^2}$

$\therefore \int \frac{2u^2}{9+u^2}\mathrm{d}u = \int \left(2 - \frac{18}{9+u^2}\right)\mathrm{d}u$

$= \int \left(2 - \frac{18}{9\left(1+\frac{1}{9}u^2\right)}\right)\mathrm{d}u$

$= \int \left(2 - \frac{2}{1+\left(\frac{u}{3}\right)^2}\right)\mathrm{d}u$

$= 2u - 2 \times \frac{1}{\frac{1}{3}} \times \arctan\left(\frac{u}{3}\right) + c$

$= 2u - 6\arctan\left(\frac{u}{3}\right) + c$

$= 2\sqrt{x-3} - 6\arctan\left(\frac{\sqrt{x-3}}{3}\right) + c$

17)

Use Integration by Substitution:

$u^2 = x + 1 \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(u^2) = \frac{\mathrm{d}}{\mathrm{d}x}(x+1) \Rightarrow 2u\frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

$x + 10 = u^2 + 9$

$\int \frac{x}{\sqrt{x+1}(x+10)}\mathrm{d}x = \int \frac{u^2-1}{u(u^2+9)} \times 2u\,\mathrm{d}u = \int \frac{2u^2-2}{u^2+9}\mathrm{d}u$

$\frac{2u^2-2}{u^2+9} \equiv \frac{2u^2+18-20}{u^2+9} \equiv \frac{2u^2+18}{u^2+9} - \frac{20}{u^2+9} \equiv \frac{2(u^2+9)}{u^2+9} - \frac{20}{u^2+9} \equiv 2 - \frac{20}{9+u^2}$

Or using polynomial division:

	2	
u^2	$2u^2$	$0u$
$+0u$	$0u$	-20
$+9$	18	

$\therefore \frac{2u^2-2}{u^2+9} \equiv 2 - \frac{20}{9+u^2}$

$\therefore \int \frac{2u^2-2}{u^2+9}\mathrm{d}u = \int \left(2 - \frac{20}{9+u^2}\right)\mathrm{d}u$

$= \int \left(2 - \frac{20}{9\left(1+\frac{1}{9}u^2\right)}\right)\mathrm{d}u$

$= 2u - \frac{20}{9} \times \frac{1}{\frac{1}{3}} \times \arctan\left(\frac{u}{3}\right) + c$

$= 2u - \frac{20}{3}\arctan\left(\frac{u}{3}\right) + c$

$= 2\sqrt{x+1} - \frac{20}{3}\arctan\left(\frac{\sqrt{x+1}}{3}\right) + c$

18)

$\int_{-3}^{-1} \frac{\sqrt{x+4}}{x+7}\mathrm{d}x$

Use Integration by Substitution:

Let $u^2 = x + 4$ where $u > 0$

$u^2 = x + 4 \Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(u^2) = \frac{\mathrm{d}}{\mathrm{d}x}(x+4) \Rightarrow 2u\frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

$x + 7 = u^2 + 3$

Change Limits:

$x = -1 \Rightarrow u^2 = 3 \Rightarrow u = \sqrt{3}$

$x = -3 \Rightarrow u^2 = 1 \Rightarrow u = 1$

$\therefore \int_{-3}^{-1} \frac{\sqrt{x+4}}{x+7}\mathrm{d}x = \int_{1}^{\sqrt{3}} \frac{\sqrt{u^2}}{u^2+3} \times 2u\,\mathrm{d}u = \int_{1}^{\sqrt{3}} \frac{2u^2}{3+u^2}\mathrm{d}u$

$\frac{2u^2}{3+u^2} \equiv \frac{2u^2+6-6}{3+u^2} \equiv \frac{6+2u^2}{3+u^2} - \frac{6}{3+u^2} \equiv \frac{2(3+u^2)}{3+u^2} - \frac{6}{3+u^2} \equiv 2 - \frac{6}{3+u^2}$

Or using polynomial division:

	2	
u^2	$2u^2$	$0u$
$+0u$	$0u$	-6
$+3$	6	

$\therefore \frac{2u^2}{3+u^2} \equiv 2 - \frac{6}{3+u^2}$

$\therefore \int_1^{\sqrt{3}} \frac{2u^2}{3+u^2}\,\mathrm{d}u = \int_1^{\sqrt{3}} \left(2 - \frac{6}{3+u^2}\right)\mathrm{d}u$

$= \int_1^{\sqrt{3}} \left(2 - \frac{6}{3\left(1+\frac{1}{3}u^2\right)}\right)\mathrm{d}u$

$= \int_1^{\sqrt{3}} \left(2 - \frac{2}{1+\left(\frac{u}{\sqrt{3}}\right)^2}\right)\mathrm{d}u$

$= \left[2u - 2 \times \sqrt{3} \times \arctan\left(\frac{u}{\sqrt{3}}\right)\right]_1^{\sqrt{3}}$

$= \left[2u - 2\sqrt{3}\arctan\left(\frac{u}{\sqrt{3}}\right)\right]_1^{\sqrt{3}}$

$= \left(2\sqrt{3} - 2\sqrt{3}\arctan 1\right) - \left(2 - 2\sqrt{3}\arctan\left(\frac{1}{\sqrt{3}}\right)\right)$

$= \left(2\sqrt{3} - 2\sqrt{3} \times \frac{\pi}{4}\right) - \left(2 - 2\sqrt{3} \times \frac{\pi}{6}\right)$

$= 2\sqrt{3} - 2 + \left(\frac{\sqrt{3}}{3} - \frac{\sqrt{3}}{2}\right)\pi$

$= 2\sqrt{3} - 2 + \frac{\pi\sqrt{3}}{6}$

$= 2\sqrt{3} - 2 + \frac{3\pi}{6\sqrt{3}}$

$= 2\sqrt{3} - 2 + \frac{\pi}{2\sqrt{3}}$ units^2 as required

19) a)

$\int \arcsin x \, \mathrm{d}x = \int 1 \times \arcsin x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arcsin x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}} \qquad v = x$

$\therefore \int \arcsin x \, \mathrm{d}x = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$

For $\int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x = \int \frac{x}{\sqrt{u}} \times -\frac{1}{2x} \mathrm{d}u = -\frac{1}{2} \int u^{-\frac{1}{2}} \mathrm{d}u = -u^{\frac{1}{2}} + c = -\sqrt{1-x^2} + c$

So:

$\int \arcsin x \, \mathrm{d}x = x \arcsin x + \sqrt{1-x^2} + c$

19) b)

$\int \arccos x \, \mathrm{d}x = \int 1 \times \arccos x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arccos x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{\sqrt{1-x^2}} \qquad v = x$

$\therefore \int \arccos x \, \mathrm{d}x = x \arccos x + \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$

For $\int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x = \int \frac{x}{\sqrt{u}} \times -\frac{1}{2x} \mathrm{d}u = -\frac{1}{2} \int u^{-\frac{1}{2}} \mathrm{d}u = -u^{\frac{1}{2}} + c = -\sqrt{1-x^2} + c$

So:

$\int \arccos x \, \mathrm{d}x = x \arccos x - \sqrt{1-x^2} + c$

19) c)

$\int \arctan x \, \mathrm{d}x = \int 1 \times \arctan x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arctan x \qquad \dfrac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1}{1+x^2} \qquad v = x$

$\therefore \int \arctan x \, \mathrm{d}x = x \arctan x - \int \frac{x}{1+x^2} \mathrm{d}x$

For $\int \frac{x}{1+x^2} \mathrm{d}x$ use Integration by Substitution (or use $\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \mathrm{d}x = |\mathrm{f}(x)| + c$):

Let $u = 1 + x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x$

$\Rightarrow \mathrm{d}x = \frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{1+x^2} \mathrm{d}x = \int \frac{x}{u} \times \frac{1}{2x} \mathrm{d}u = \frac{1}{2} \int \frac{1}{u} \mathrm{d}u = \frac{1}{2} \ln|u| + c = \frac{1}{2} \ln|1 + x^2| + c$

So:

$\int \arctan x \, \mathrm{d}x = x \arctan x - \frac{1}{2} \ln|1 + x^2| + c$

20)

$\int_0^1 8\arctan x\,\mathrm{d}x$

Use Integration by Parts:

$u = \arctan x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = 8$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2}$ $\qquad$ $v = 8x$

$\therefore \int 8\arctan x\,\mathrm{d}x = 8x\arctan x - \int \frac{8x}{1+x^2}\mathrm{d}x$

For $\int \frac{8x}{1+x^2}\mathrm{d}x$ use Integration by Substitution (or use $\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)}\mathrm{d}x = |\mathrm{f}(x)| + c$):

Let $u = 1 + x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x$

$\Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$\therefore \int \frac{8x}{1+x^2}\mathrm{d}x = \int \frac{8x}{u} \times \frac{1}{2x}\mathrm{d}u = 4\int \frac{1}{u}\mathrm{d}u = 4\ln|u| + c = 4\ln|1+x^2| + c$

So:

$\int_0^1 8\arctan x\,\mathrm{d}x = [8x\arctan x + 4\ln|1+x^2|]_0^1$

$= (8 \times 1 \times \arctan 1 + 4\ln|1+1^2|) - (8 \times 0 \times \arctan 0 + 4\ln|1+0^2|)$

$= (2\pi + 4\ln 2) - 0$

$= 2\pi + \ln 16$

$\therefore p = 16$

21) a)

$\int \arcsin 2x \, \mathrm{d}x = \int 1 \times \arcsin 2x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arcsin 2x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{\sqrt{1-4x^2}} \qquad v = x$

$\therefore \int \arcsin 2x \, \mathrm{d}x = x \arcsin 2x - \int \frac{2x}{\sqrt{1-4x^2}} \mathrm{d}x$

For $\int \frac{2x}{\sqrt{1-4x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - 4x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -8x$

$\Rightarrow \mathrm{d}x = -\frac{1}{8x} \mathrm{d}u$

$\therefore \int \frac{2x}{\sqrt{1-4x^2}} \mathrm{d}x = \int \frac{2x}{\sqrt{u}} \times -\frac{1}{8x} \mathrm{d}u = -\frac{1}{4} \int u^{-\frac{1}{2}} \mathrm{d}u = -\frac{1}{2} u^{\frac{1}{2}} + c = -\frac{1}{2}\sqrt{1-4x^2} + c$

So:

$\int \arcsin 2x \, \mathrm{d}x = x \arcsin 2x + \frac{1}{2}\sqrt{1-4x^2} + c$

21) b)

$\int \arccos 3x \, \mathrm{d}x = \int 1 \times \arccos 3x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arccos 3x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{3}{\sqrt{1-9x^2}} \qquad v = x$

$\therefore \int \arccos 3x \, \mathrm{d}x = x \arccos 3x + \int \frac{3x}{\sqrt{1-9x^2}} \mathrm{d}x$

For $\int \frac{3x}{\sqrt{1-9x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - 9x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -18x$

$\Rightarrow \mathrm{d}x = -\frac{1}{18x} \mathrm{d}u$

$\therefore \int \frac{3x}{\sqrt{1-9x^2}} \mathrm{d}x = \int \frac{3x}{\sqrt{u}} \times -\frac{1}{18x} \mathrm{d}u = -\frac{1}{6} \int u^{-\frac{1}{2}} \mathrm{d}u = -\frac{1}{3} u^{\frac{1}{2}} + c = -\frac{1}{3}\sqrt{1-9x^2} + c$

So:

$\int \arccos 3x \, \mathrm{d}x = x \arccos 3x - \frac{1}{3}\sqrt{1-9x^2} + c$

21) c)

$\int \arctan 4x \, \mathrm{d}x = \int 1 \times \arctan 4x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arctan 4x \qquad \dfrac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{4}{1+16x^2} \qquad v = x$

$\therefore \int \arctan 4x \, \mathrm{d}x = x \arctan 4x - \int \frac{4x}{1+16x^2} \mathrm{d}x$

For $\int \frac{4x}{1+16x^2} \mathrm{d}x$ use Integration by Substitution (or use $\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \mathrm{d}x = |\mathrm{f}(x)| + c$):

Let $u = 1 + 16x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 32x$

$\Rightarrow \mathrm{d}x = \frac{1}{32x} \mathrm{d}u$

$\therefore \int \frac{4x}{1+16x^2} \mathrm{d}x = \int \frac{4x}{u} \times \frac{1}{32x} \mathrm{d}u = \frac{1}{8} \int \frac{1}{u} \mathrm{d}u = \frac{1}{8} \ln|u| + c = \frac{1}{8} \ln|1 + 16x^2| + c$

So:

$\int \arctan 4x \, \mathrm{d}x = x \arctan 4x - \frac{1}{8} \ln|1 + 16x^2| + c$

22) a)

$\int \arcsin x \, \mathrm{d}x = \int 1 \times \arcsin x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arcsin x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}} \qquad v = x$

$\therefore \int \arcsin x \, \mathrm{d}x = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$

For $\int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x = \int \frac{x}{\sqrt{u}} \times -\frac{1}{2x} \mathrm{d}u = -\frac{1}{2} \int u^{-\frac{1}{2}} \mathrm{d}u = -u^{\frac{1}{2}} + c = -\sqrt{1-x^2} + c$

So:

$\int \arcsin x \, \mathrm{d}x = x \arcsin x + \sqrt{1-x^2} + c$

So by reversing the chain rule:

$\int 8x \arcsin(x^2) \, \mathrm{d}x$

$= 4\left(x^2 \arcsin(x^2) + \sqrt{1-(x^2)^2}\right) + c$

$= 4x^2 \arcsin(x^2) + 4\sqrt{1-x^4} + c$

22) b)

$\int \arccos x \, \mathrm{d}x = \int 1 \times \arccos x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arccos x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{\sqrt{1-x^2}} \qquad v = x$

$\therefore \int \arccos x \, \mathrm{d}x = x \arccos x + \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$

For $\int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x = \int \frac{x}{\sqrt{u}} \times -\frac{1}{2x} \mathrm{d}u = -\frac{1}{2} \int u^{-\frac{1}{2}} \mathrm{d}u = -u^{\frac{1}{2}} + c = -\sqrt{1-x^2} + c$

So:

$\int \arccos x \, \mathrm{d}x = x \arccos x - \sqrt{1-x^2} + c$

So by reversing the chain rule:

$\int \mathrm{e}^{2x} \arccos(\mathrm{e}^{2x}) \, \mathrm{d}x$

$= \frac{1}{2}\left(\mathrm{e}^{2x} \arccos(\mathrm{e}^{2x}) - \sqrt{1-(\mathrm{e}^{2x})^2}\right) + c$

$= \frac{1}{2}\mathrm{e}^{2x} \arccos(\mathrm{e}^{2x}) - \frac{1}{2}\sqrt{1-\mathrm{e}^{4x}} + c$

22) c)

$\int \arctan x \, \mathrm{d}x = \int 1 \times \arctan x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arctan x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2} \qquad v = x$

$\therefore \int \arctan x \, \mathrm{d}x = x \arctan x - \int \frac{x}{1+x^2} \mathrm{d}x$

For $\int \frac{x}{1+x^2} \mathrm{d}x$ use Integration by Substitution (or use $\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \mathrm{d}x = |\mathrm{f}(x)| + c$):

Let $u = 1 + x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x$

$\Rightarrow \mathrm{d}x = \frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{1+x^2} \mathrm{d}x = \int \frac{x}{u} \times \frac{1}{2x} \mathrm{d}u = \frac{1}{2} \int \frac{1}{u} \mathrm{d}u = \frac{1}{2} \ln|u| + c = \frac{1}{2} \ln|1 + x^2| + c$

So:

$\int \arctan x \, \mathrm{d}x = x \arctan x - \frac{1}{2} \ln|1 + x^2| + c$

So by reversing the chain rule:

$\int \sin x \arctan(\cos x) \, \mathrm{d}x$

$= -\left(\cos x \arctan(\cos x) - \frac{1}{2} \ln|1 + (\cos x)^2|\right) + c$

$= -\cos x \arctan(\cos x) + \frac{1}{2} \ln|1 + \cos^2 x| + c$

22) d)

$\int \arcsin x \, \mathrm{d}x = \int 1 \times \arcsin x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arcsin x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}} \qquad v = x$

$\therefore \int \arcsin x \, \mathrm{d}x = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$

For $\int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x = \int \frac{x}{\sqrt{u}} \times -\frac{1}{2x} \mathrm{d}u = -\frac{1}{2} \int u^{-\frac{1}{2}} \mathrm{d}u = -u^{\frac{1}{2}} + c = -\sqrt{1-x^2} + c$

So:

$\int \arcsin x \, \mathrm{d}x = x \arcsin x + \sqrt{1-x^2} + c$

So by reversing the chain rule:

$\int \sec^2 x \arcsin(\tan x) \, \mathrm{d}x$

$= \tan x \arcsin(\tan x) + \sqrt{1 - \tan^2 x} + c$

22) e)

$\int \arctan x \, \mathrm{d}x = \int 1 \times \arctan x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arctan x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2} \qquad v = x$

$\therefore \int \arctan x \, \mathrm{d}x = x \arctan x - \int \frac{x}{1+x^2} \mathrm{d}x$

For $\int \frac{x}{1+x^2} \mathrm{d}x$ use Integration by Substitution (or use $\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \mathrm{d}x = |\mathrm{f}(x)| + c$):

Let $u = 1 + x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x$

$\Rightarrow \mathrm{d}x = \frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{1+x^2} \mathrm{d}x = \int \frac{x}{u} \times \frac{1}{2x} \mathrm{d}u = \frac{1}{2} \int \frac{1}{u} \mathrm{d}u = \frac{1}{2} \ln|u| + c = \frac{1}{2} \ln|1 + x^2| + c$

So:

$\int \arctan x \, \mathrm{d}x = x \arctan x - \frac{1}{2} \ln|1 + x^2| + c$

So by reversing the chain rule:

$\int 3^x \arctan(3^x) \, \mathrm{d}x$

$= \frac{1}{\ln 3} \left(3^x \arctan 3^x - \frac{1}{2} \ln|1 + 3^{2x}| \right) + c$

23) a)

$\int x \arcsin x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arcsin x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}} \qquad v = \frac{1}{2}x^2$

$\therefore \int x \arcsin x \, \mathrm{d}x = \frac{1}{2}x^2 \arcsin x - \frac{1}{2}\int \frac{x^2}{\sqrt{1-x^2}} \mathrm{d}x$

For $\int \frac{x^2}{\sqrt{1-x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $x = \sin u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u$

$\Rightarrow \mathrm{d}x = \cos u \, \mathrm{d}u$

$\therefore \int \frac{x^2}{\sqrt{1-x^2}} \mathrm{d}x = \int \frac{\sin^2 u}{\sqrt{1-\sin^2 u}} \times \cos u \, \mathrm{d}u = \int \frac{\sin^2 u \cos u}{\sqrt{\cos^2 u}} \mathrm{d}u = \int \frac{\sin^2 u \cos u}{\cos u} \mathrm{d}u$

$= \int \sin^2 u \, \mathrm{d}u = \int \left(\frac{1}{2} - \frac{1}{2}\cos 2u\right) \mathrm{d}u = \frac{1}{2}u - \frac{1}{4}\sin 2u + c = \frac{1}{2}u - \frac{1}{2}\sin u \cos u + c$

$= \frac{1}{2}u - \frac{1}{2}\sin u \sqrt{1-\sin^2 u} + c = \frac{1}{2}\arcsin x - \frac{1}{2}x\sqrt{1-x^2} + c$

So:

$\int x \arcsin x \, \mathrm{d}x = \frac{1}{2}x^2 \arcsin x - \frac{1}{2}\left(\frac{1}{2}\arcsin x - \frac{1}{2}x\sqrt{1-x^2}\right) + c$

$= \frac{1}{2}x^2 \arcsin x - \frac{1}{4}\arcsin x + \frac{1}{4}x\sqrt{1-x^2} + c$

23) b)

$\int x^2 \arccos x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arccos x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x^2$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{\sqrt{1-x^2}} \qquad v = \frac{1}{3}x^3$

$\therefore \int x^2 \arccos x \, \mathrm{d}x = \frac{1}{3}x^3 \arccos x + \frac{1}{3}\int \frac{x^3}{\sqrt{1-x^2}} \mathrm{d}x$

For $\int \frac{x^3}{\sqrt{1-x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x}\mathrm{d}u$

$\therefore \int \frac{x^3}{\sqrt{1-x^2}} \mathrm{d}x = \int \frac{x^3}{\sqrt{u}} \times -\frac{1}{2x} \mathrm{d}u = -\frac{1}{2}\int \frac{x^2}{\sqrt{u}} \mathrm{d}u = -\frac{1}{2}\int \frac{1-u}{u^{\frac{1}{2}}} \mathrm{d}u = \frac{1}{2}\int \left(u^{\frac{1}{2}} - u^{-\frac{1}{2}}\right) \mathrm{d}u$

$= \frac{1}{3}u^{\frac{3}{2}} - u^{\frac{1}{2}} + c = \frac{1}{3}(1-x^2)^{\frac{3}{2}} - (1-x^2)^{\frac{1}{2}} + c$

So:

$\int x^2 \arccos x \, \mathrm{d}x = \frac{1}{3}x^3 \arccos x + \frac{1}{3}\left(\frac{1}{3}(1-x^2)^{\frac{3}{2}} - (1-x^2)^{\frac{1}{2}}\right) + c$

$= \frac{1}{3}x^3 \arccos x + \frac{1}{9}(1-x^2)^{\frac{3}{2}} - \frac{1}{3}(1-x^2)^{\frac{1}{2}} + c$

23) c)

$\int x^3 \arctan x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arctan x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x^3$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2} \qquad v = \frac{1}{4}x^4$

$\therefore \int x^3 \arctan x \, \mathrm{d}x = \frac{1}{4}x^4 \arctan x - \frac{1}{4}\int \frac{x^4}{1+x^2}\mathrm{d}x$

Using polynomial division:

	x^2	$0x$	-1	
x^2	x^4	$0x^3$	$-x^2$	
$+0x$	$0x^3$	$0x^2$	$0x$	$+1$
$+1$	x^2	$0x$	-1	

$\therefore \frac{x^4}{1+x^2} \equiv x^2 - 1 + \frac{1}{1+x^2}$

So:

$\int x^3 \arctan x \, \mathrm{d}x = \frac{1}{4}x^4 \arctan x - \frac{1}{4}\int \left(x^2 - 1 + \frac{1}{1+x^2}\right)\mathrm{d}x$

$= \frac{1}{4}x^4 \arctan x - \frac{1}{4}\left(\frac{1}{3}x^3 - x + \arctan x\right) + c$

$= \frac{1}{4}x^4 \arctan x - \frac{1}{12}x^3 + \frac{1}{4}x - \frac{1}{4}\arctan x + c$

24) a)

$\frac{10x^2}{1+x^2} \equiv \frac{10x^2+10-10}{1+x^2} \equiv \frac{10x^2+10}{1+x^2} - \frac{10}{1+x^2} \equiv \frac{10(1+x^2)}{1+x^2} - \frac{10}{1+x^2} \equiv 10 - \frac{10}{1+x^2}$

Or using polynomial division:

	10	
x^2	$10x^2$	$0x$
$+0x$	$0x$	-10
$+1$	10	

$\therefore \frac{10x^2}{1+x^2} \equiv 10 - \frac{10}{1+x^2}$

24) b)

$\int 20x \arctan x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arctan x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 20x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2} \qquad v = 10x^2$

$\therefore \int 20x \arctan x \, \mathrm{d}x = 10x^2 \arctan x - \int \frac{10x^2}{1+x^2} \mathrm{d}x$

Consider $\int \frac{10x^2}{1+x^2} \mathrm{d}x = \int \left(10 - \frac{10}{1+x^2}\right) \mathrm{d}x$

$= 10x - 10 \arctan x + c$

So:

$\int 20x \arctan x \, \mathrm{d}x = 10x^2 \arctan x - (10x - 10 \arctan x) + c$

$= 10x^2 \arctan x - 10x + 10 \arctan x + c$

25) a)

$\frac{15x^2}{4+x^2} \equiv \frac{15x^2+60-60}{4+x^2} \equiv \frac{15x^2+60}{4+x^2} - \frac{60}{4+x^2} \equiv \frac{15(4+x^2)}{4+x^2} - \frac{60}{4+x^2} \equiv 15 - \frac{60}{4+x^2}$

Or using polynomial division:

	15	
x^2	$15x^2$	$0x$
$+0x$	$0x$	-60
$+4$	60	

$\therefore \frac{15x^2}{4+x^2} \equiv 15 - \frac{60}{4+x^2}$

25) b)

$\int 30x \arctan\left(\frac{x}{2}\right) \mathrm{d}x$

Use Integration by Parts:

$u = \arctan\left(\frac{x}{2}\right)$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = 30x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{4+x^2}$ $\qquad$ $v = 15x^2$

Note that $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+\left(\frac{x}{2}\right)^2} \times \frac{1}{2} = \frac{1}{2+\frac{2}{4}x^2} = \frac{2}{4+x^2}$

$\therefore \int 30x \arctan\left(\frac{x}{2}\right) \mathrm{d}x = 15x^2 \arctan\left(\frac{x}{2}\right) - \int \frac{30x^2}{4+x^2} \mathrm{d}x$

Consider $\int \frac{30x^2}{4+x^2} \mathrm{d}x = \int \left(30 - \frac{120}{4+x^2}\right) \mathrm{d}x$

$= 30x - 60 \arctan\left(\frac{x}{2}\right) + c$

So:

$\int 30x \arctan\left(\frac{x}{2}\right) \mathrm{d}x = 15x^2 \arctan\left(\frac{x}{2}\right) - \left(30x - 60 \arctan\left(\frac{x}{2}\right)\right) + c$

$= 15x^2 \arctan\left(\frac{x}{2}\right) - 30x + 60 \arctan\left(\frac{x}{2}\right) + c$

26)

$\int_0^1 6x \arcsin(x^2)\,\mathrm{d}x$

Use Integration by Substitution

Let $u = x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x$

$\Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

Change Limits:

$x = 1 \Rightarrow u = 1^2 = 1$

$x = 0 \Rightarrow u = 0^2 = 0$

$\therefore \int_0^1 6x \arcsin(x^2)\,\mathrm{d}x = \int_0^1 6x \arcsin u \times \frac{1}{2x}\mathrm{d}u = 3\int_0^1 \arcsin u\,\mathrm{d}u$

Consider $\int \arcsin x\,\mathrm{d}x = \int 1 \times \arcsin x\,\mathrm{d}x$ and use Integration by Parts:

$u = \arcsin x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}} \qquad v = x$

$\therefore \int \arcsin x\,\mathrm{d}x = x\arcsin x - \int \frac{x}{\sqrt{1-x^2}}\mathrm{d}x$

For $\int \frac{x}{\sqrt{1-x^2}}\mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x}\mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{1-x^2}}\mathrm{d}x = \int \frac{x}{\sqrt{u}} \times -\frac{1}{2x}\mathrm{d}u = -\frac{1}{2}\int u^{-\frac{1}{2}}\mathrm{d}u = -u^{\frac{1}{2}} + c = -\sqrt{1-x^2} + c$

So:

$\int \arcsin x\,\mathrm{d}x = x\arcsin x + \sqrt{1-x^2} + c$

So now returning to the original integral:

$\int_0^1 6x \arcsin(x^2)\,\mathrm{d}x = 3\int_0^1 \arcsin u\,\mathrm{d}u = 3\left[u\arcsin u + \sqrt{1-u^2}\right]_0^1$

$= 3\left(\left(1 \times \arcsin 1 + \sqrt{1-1^2}\right) - \left(0 \times \arcsin 0 + \sqrt{1-0^2}\right)\right)$

$= 3\left(\left(\frac{\pi}{2} + 0\right) - (0+1)\right)$

$= 3\left(\frac{\pi}{2} - 1\right)$

So the shaded area is $\left(\frac{3\pi}{2} - 3\right)$ units^2

27) a)

$\int \frac{\arcsin x}{\sqrt{1-x^2}}\,\mathrm{d}x$

Use Integration by Parts:

$u = \arcsin x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}} \qquad v = \arcsin x$

$\therefore \int \frac{\arcsin x}{\sqrt{1-x^2}}\,\mathrm{d}x = (\arcsin x)^2 - \int \frac{\arcsin x}{\sqrt{1-x^2}}\,\mathrm{d}x$

$\Rightarrow 2\int \frac{\arcsin x}{\sqrt{1-x^2}}\,\mathrm{d}x = (\arcsin x)^2$

$\therefore \int \frac{\arcsin x}{\sqrt{1-x^2}}\,\mathrm{d}x = \frac{1}{2}(\arcsin x)^2 + c$

Alternatively:

Let $u = \arcsin x$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}$

$\Rightarrow \mathrm{d}x = \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}u$

$\therefore \int \frac{\arcsin x}{\sqrt{1-x^2}}\,\mathrm{d}x = \int u\,\mathrm{d}u = \frac{1}{2}u^2 + c$

$= \frac{1}{2}(\arcsin x)^2 + c$

27) b)

$\int \frac{\arccos x}{\sqrt{1-x^2}}\,\mathrm{d}x$

Use Integration by Parts:

$u = \arccos x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{\sqrt{1-x^2}} \qquad v = -\arccos x$

$\therefore \int \frac{\arccos x}{\sqrt{1-x^2}}\,\mathrm{d}x = -(\arccos x)^2 - \int \frac{\arccos x}{\sqrt{1-x^2}}\,\mathrm{d}x$

$\Rightarrow 2\int \frac{\arccos x}{\sqrt{1-x^2}}\,\mathrm{d}x = -(\arccos x)^2$

$\therefore \int \frac{\arccos x}{\sqrt{1-x^2}}\,\mathrm{d}x = -\frac{1}{2}(\arccos x)^2 + c$

Alternatively:

Let $u = \arccos x$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{\sqrt{1-x^2}}$

$\Rightarrow \mathrm{d}x = -\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}u$

$\therefore \int \frac{\arccos x}{\sqrt{1-x^2}}\,\mathrm{d}x = -\int u\,\mathrm{d}u = -\frac{1}{2}u^2 + c$

$= -\frac{1}{2}(\arccos x)^2 + c$

27) c)

$\int \frac{\arctan x}{1+x^2}\,\mathrm{d}x$

Use Integration by Parts:

$u = \arctan x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{1+x^2}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2} \qquad v = \arctan x$

$\therefore \int \frac{\arctan x}{1+x^2}\,\mathrm{d}x = (\arctan x)^2 - \int \frac{\arctan x}{1+x^2}\,\mathrm{d}x$

$\Rightarrow 2\int \frac{\arctan x}{1+x^2}\,\mathrm{d}x = (\arctan x)^2$

$\therefore \int \frac{\arctan x}{1+x^2}\,\mathrm{d}x = \frac{1}{2}(\arctan x)^2 + c$

Alternatively:

Let $u = \arctan x$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2}$

$\Rightarrow \mathrm{d}x = \frac{1}{1+x^2}\,\mathrm{d}u$

$\therefore \int \frac{\arctan x}{1+x^2}\,\mathrm{d}x = \int u\,\mathrm{d}u = \frac{1}{2}u^2 + c$

$= \frac{1}{2}(\arctan x)^2 + c$

28)

$\int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,\mathrm{d}x$

Use Integration by Parts:

$u = (\arcsin x)^2 \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2\arcsin x}{\sqrt{1-x^2}} \qquad v = \arcsin x$

$\therefore \int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,\mathrm{d}x = [(\arcsin x)^3]_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} - 2\int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,\mathrm{d}x$

$\Rightarrow 3\int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,\mathrm{d}x = [(\arcsin x)^3]_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}}$

$\Rightarrow \int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,\mathrm{d}x = \frac{1}{3}[(\arcsin x)^3]_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}}$

$= \frac{1}{3}\left(\left(\arcsin\frac{1}{\sqrt{2}}\right)^3 - \left(\arcsin\frac{1}{2}\right)^3\right)$

$= \frac{1}{3}\left(\left(\frac{\pi}{4}\right)^3 - \left(\frac{\pi}{6}\right)^3\right)$

$= \frac{1}{3}\left(\frac{\pi^3}{64} - \frac{\pi^3}{216}\right)$

$= \frac{19\pi^3}{5184}$

So $k = \frac{19}{5184}$

Alternatively:

Let $u = \arcsin x$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}} \Rightarrow \mathrm{d}x = \sqrt{1-x^2}\mathrm{d}u$

When $x = \frac{1}{2}$, $u = \arcsin\frac{1}{2} = \frac{\pi}{6}$

When $x = \frac{1}{\sqrt{2}}$, $u = \arcsin\frac{1}{\sqrt{2}} = \frac{\pi}{4}$

$\int_{\frac{1}{2}}^{\frac{1}{\sqrt{2}}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,\mathrm{d}x = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} u^2\,\mathrm{d}u = \left[\frac{1}{3}u^3\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$

$= \frac{1}{3}\left(\frac{\pi}{4}\right)^3 - \frac{1}{3}\left(\frac{\pi}{6}\right)^3$

$= \frac{\pi^3}{192} - \frac{\pi^3}{648}$

$= \frac{19\pi^3}{5184}$

So $k = \frac{19}{5184}$

29)

$\int \arctan x \ln(1+x^2)\,\mathrm{d}x$

Use Integration by Parts:

$u = \arctan x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \ln(1+x^2)$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2} \qquad v = ?$

In order to integrate $\ln(1+x^2)$ we must use integration by parts.

$\int \ln(1+x^2)\,\mathrm{d}x = \int 1 \times \ln(1+x^2)\,\mathrm{d}x$

$u = \ln(1+x^2) \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2x}{1+x^2} \qquad v = x$

$\therefore \int \ln(1+x^2)\,\mathrm{d}x = x\ln(1+x^2) - \int \frac{2x^2}{1+x^2}\,\mathrm{d}x$

$= x\ln(1+x^2) - \int \frac{2(1+x^2)-2}{1+x^2}\,\mathrm{d}x$

$= x\ln(1+x^2) - \int \left(2 - \frac{2}{1+x^2}\right)\mathrm{d}x$

$= x\ln(1+x^2) - (2x - 2\arctan x) + c$

$= x\ln(1+x^2) - 2x + 2\arctan x + c$

Returning to the original integral:

$\int \arctan x \ln(1+x^2)\,\mathrm{d}x$

Use Integration by Parts:

$u = \arctan x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \ln(1+x^2)$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2} \qquad v = x\ln(1+x^2) - 2x + 2\arctan x$

$\therefore \int \arctan x \ln(1+x^2)\,\mathrm{d}x = (x\ln(1+x^2) - 2x + 2\arctan x)\arctan x -$

$\int \frac{x\ln(1+x^2) - 2x + 2\arctan x}{1+x^2}\,\mathrm{d}x$

$= (x\ln(1+x^2) - 2x + 2\arctan x)\arctan x - \int \frac{x\ln(1+x^2)}{1+x^2}\,\mathrm{d}x + \int \frac{2x}{1+x^2}\,\mathrm{d}x - \int \frac{2\arctan x}{1+x^2}\,\mathrm{d}x$

1st integral: $\int \frac{x\ln(1+x^2)}{1+x^2}\,\mathrm{d}x$

Use Integration by Substitution (you can also reverse the chain rule)

Let $u = 1+x^2 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$\therefore \int \frac{x\ln(1+x^2)}{1+x^2}\,\mathrm{d}x = \int \frac{x\ln u}{u} \times \frac{1}{2x}\,\mathrm{d}u = \frac{1}{2}\int \frac{\ln u}{u}\,\mathrm{d}u = \frac{1}{2} \times \frac{1}{2}(\ln u)^2 + c = \frac{1}{4}(\ln(1+x^2))^2 + c$

2nd integral: $\int \frac{2x}{1+x^2}\,\mathrm{d}x$

Use Integration by Substitution (you can also reverse the chain rule)

Let $u = 1 + x^2 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$\therefore \int \frac{2x}{1+x^2}\mathrm{d}x = \int \frac{2x}{u} \times \frac{1}{2x}\mathrm{d}u = \int \frac{1}{u}\mathrm{d}u = \ln|u| + c = \ln(1 + x^2) + c$

3rd integral: $\int \frac{2\arctan x}{1+x^2}\mathrm{d}x$

Use Integration by Substitution (you can also reverse the chain rule)

Let $u = \arctan x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2} \Rightarrow \mathrm{d}x = (1 + x^2)\mathrm{d}u$

$\therefore \int \frac{2\arctan x}{1+x^2}\mathrm{d}x = \int \frac{2u}{1+x^2} \times (1 + x^2)\,\mathrm{d}u = \int 2u\,\mathrm{d}u = u^2 + c = (\arctan x)^2 + c$

Returning to the original integral:

$\therefore \int \arctan x \ln(1 + x^2)\,\mathrm{d}x = (x\ln(1 + x^2) - 2x + 2\arctan x)\arctan x -$

$\frac{1}{4}(\ln(1 + x^2))^2 + \ln(1 + x^2) - (\arctan x)^2 + c$

30) a)

$\int \operatorname{arccosec} x\,\mathrm{d}x = \int 1 \times \operatorname{arccosec} x\,\mathrm{d}x$

In order to use Integration by Parts, we must find the derivative of $\operatorname{arccosec} x$

Let $y = \operatorname{arccosec} x \Rightarrow \operatorname{cosec} y = x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{cosec} y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow -\operatorname{cosec} y \cot y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\operatorname{cosec} y \cot y}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\operatorname{cosec} y\sqrt{\operatorname{cosec}^2 y - 1}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{x\sqrt{x^2-1}}$

Now use Integration by Parts:

$u = \operatorname{arccosec} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{x\sqrt{x^2-1}} \qquad v = x$

$\therefore \int \operatorname{arccosec} x\,\mathrm{d}x = x\operatorname{arccosec} x + \int \frac{1}{\sqrt{x^2-1}}\mathrm{d}x$

$= x\operatorname{arccosec} x + \operatorname{arcosh} x + c$

30) b)

$\int \operatorname{arcsec} x \, \mathrm{d}x = \int 1 \times \operatorname{arcsec} x \, \mathrm{d}x$

In order to use Integration by Parts, we must find the derivative of $\operatorname{arccosec} x$

Let $y = \operatorname{arcsec} x \Rightarrow \sec y = x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\sec y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow \sec y \tan y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sec y \tan y}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sec y \sqrt{\sec^2 y - 1}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x\sqrt{x^2 - 1}}$

Now use Integration by Parts:

$u = \operatorname{arcsec} x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x\sqrt{x^2 - 1}}$ $\qquad$ $v = x$

$\therefore \int \operatorname{arcsec} x \, \mathrm{d}x = x \operatorname{arcsec} x - \int \frac{1}{\sqrt{x^2 - 1}} \mathrm{d}x$

$= x \operatorname{arcsec} x - \operatorname{arcosh} x + c$

30) c)

$\int \operatorname{arccot} x \, \mathrm{d}x = \int 1 \times \operatorname{arccot} x \, \mathrm{d}x$

In order to use Integration by Parts, we must find the derivative of $\operatorname{arccosec} x$

Let $y = \operatorname{arccot} x \Rightarrow \cot y = x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\cot y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow -\operatorname{cosec}^2 y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\operatorname{cosec}^2 y}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\cot^2 y + 1}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{x^2+1}$

Now use Integration by Parts:

$u = \operatorname{arccot} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{x^2+1} \qquad v = x$

$\therefore \int \operatorname{arccot} x \, \mathrm{d}x = x \operatorname{arccot} x + \int \frac{x}{x^2+1} \mathrm{d}x$

$= x \operatorname{arccot} x + \frac{1}{2}\ln|x^2+1| + c$

31)

$\int \operatorname{arccosec} 2x \, \mathrm{d}x = \int 1 \times \operatorname{arccosec} 2x \, \mathrm{d}x$

In order to use Integration by Parts, we must find the derivative of $\operatorname{arccosec} 2x$

Let $y = \operatorname{arccosec} 2x \Rightarrow \operatorname{cosec} y = 2x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{cosec} y) = \frac{\mathrm{d}}{\mathrm{d}x}(2x)$

$\Rightarrow -\operatorname{cosec} y \cot y \frac{\mathrm{d}y}{\mathrm{d}x} = 2$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2}{\operatorname{cosec} y \cot y}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2}{\operatorname{cosec} y\sqrt{\operatorname{cosec}^2 y - 1}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2}{2x\sqrt{(2x)^2 - 1}}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{x\sqrt{4x^2 - 1}}$

Now use Integration by Parts:

$u = \operatorname{arccosec} 2x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{x\sqrt{4x^2 - 1}} \qquad v = x$

$\therefore \int \operatorname{arccosec} 2x \, \mathrm{d}x = x \operatorname{arccosec} 2x + \int \frac{1}{\sqrt{4x^2 - 1}} \mathrm{d}x$

Now $\int \frac{1}{\sqrt{4x^2 - 1}} \mathrm{d}x = \int \frac{1}{\sqrt{(2x)^2 - 1}} \mathrm{d}x$

$= \frac{1}{2} \operatorname{arcosh} 2x + c$

$\therefore \int \operatorname{arccosec} 2x \, \mathrm{d}x = x \operatorname{arccosec} 2x + \frac{1}{2} \operatorname{arcosh} 2x + c$

$\therefore \int_{\frac{1}{\sqrt{2}}}^{1} \operatorname{arccosec} 2x \ \mathrm{d}x = \left[x \operatorname{arccosec} 2x + \frac{1}{2} \operatorname{arcosh} 2x \right]_{\frac{1}{\sqrt{2}}}^{1}$

$= \left(\operatorname{arccosec} 2 + \frac{1}{2} \operatorname{arcosh} 2 \right) - \left(\frac{1}{\sqrt{2}} \operatorname{arccosec} \left(\frac{2}{\sqrt{2}} \right) + \frac{1}{2} \operatorname{arcosh} \left(\frac{2}{\sqrt{2}} \right) \right)$

$= \left(\frac{\pi}{6} + \frac{1}{2} \ln\left(2 + \sqrt{3}\right) \right) - \left(\frac{\pi}{4\sqrt{2}} + \frac{1}{2} \ln\left(\frac{2}{\sqrt{2}} + \sqrt{\left(\frac{2}{\sqrt{2}} \right)^2 - 1} \right) \right)$

$= \left(\frac{\pi}{6} + \frac{1}{2} \ln\left(2 + \sqrt{3}\right) \right) - \left(\frac{\pi}{4\sqrt{2}} + \frac{1}{2} \ln\left(1 + \sqrt{2}\right) \right)$

$= \left(\frac{1}{2} \ln\left(\frac{2+\sqrt{3}}{1+\sqrt{2}} \right) + \frac{\pi}{6} - \frac{\pi}{4\sqrt{2}} \right)$ units^2 as req.